



Society of Petroleum Engineers

SPE-227078-MS

Core/Shell Quantum Dots: Highly Stable and Tunable Tracers for Oil Reservoir Applications

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227078-MS](https://doi.org/10.2118/227078-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Monitoring fluid behavior and interactions within petroleum reservoirs relies heavily on effective tracer technologies. However, conventional tracers, including certain nanomaterials like carbon dots, often present challenges such as inflexible emissive characteristics and insufficient stability under the demanding conditions typical of oil reservoirs. This paper introduces an innovative core/shell quantum dot (QD) system designed to overcome these limitations. The developed QDs offer the potential for tunable emission and demonstrate exceptional stability across a range of rigorous experimental conditions. These nanomaterials are synthesized with relative ease and exhibit significant resilience when exposed to water, saline solutions (e.g. NaCl), and diverse crude oil compositions. Critically, their fluorescent emission properties show minimal alteration when subjected to variations in pressure and temperature. This new class of QDs signifies a substantial advancement in tracer technology, offering exceptionally stable and potentially customizable emissive agents for oilfield applications. They possess considerable potential for identifying fluid sources and investigating chemical and physical transformations during upstream oil and gas operations, thereby contributing to enhanced reservoir management and monitoring strategies.

Introduction

Tracer technologies are indispensable in the oil and gas industry, providing critical data for understanding fluid movement, assessing connectivity between wells, and characterizing reservoir heterogeneity.(Al-Qasim et al. 2020, Sanni et al. 2018, Haugan et al. 2013, Haugan and Axelsson 2013, Sun et al. 2023) These methods offer valuable insights into the dynamics of subsurface fluids, including breakthrough times and the efficiency of sweep operations during oil recovery processes. For effective tracing in crude oil environments, materials that emit in the red spectral region and possess long-term stability are highly desirable. The inherent stability of quantum dots (QDs), particularly concerning their excited-state properties, alongside their resilience in aqueous, saline, and varied oil matrixes, are paramount considerations.(Franco et al. 2020, Kumari and Ahmaruzzaman 2023, Giroux et al. 2022, Cotta 2020, Cheng et al. 2021, Hu and Zhu

2015, Al-Qasim et al. 2020, Hu et al. 2019, Sanni et al. 2018, Alaskar and Kosynkin 2017) However, the application of tracers is not without challenges. For instance, tracers can adsorb onto rock surfaces, potentially leading to inaccurate interpretations of flow paths. Furthermore, the dilute concentrations of tracers in large reservoir volumes necessitate highly sensitive detection equipment, which can escalate operational costs. A fundamental requirement is that tracers must remain inert, neither reacting with reservoir fluids nor altering the existing reservoir chemistry.

Quantum dots have emerged as promising candidates for tracer applications due to their unique advantages, such as straightforward synthesis routes, favorable environmental profiles, and outstanding photophysical and chemical properties. (Franco et al. 2020, Bera et al. 2010, Medina et al. 2024, Izadi and Nasernejad 2022, Roy et al. 2023, Pan et al. 2019) Their characteristics, including high crystallinity, excellent dispersibility, low toxicity, and distinctive photoluminescence, position QDs as a technology capable of enhancing various oil and gas operations. While some nanomaterials, like carbon quantum dots (CQDs), have shown promise with good stability in extreme conditions and mobility in porous media, their luminescent characteristics are not always optimal for oil exploration applications. Although combining partitioning and conservative tracers can aid in estimating non-aqueous-phase liquid distribution in porous media, such approaches are often limited by the instability of the tracers in harsh reservoir conditions. Despite the widespread application of QDs in other scientific and technological domains, comprehensive studies on their specific utility and impact within the oil and gas sector, particularly concerning large-scale environmental effects, remain relatively scarce. Previous research has examined the stability of various QDs, including water-soluble CdSe QDs that showed aggregation in seawater and surface-modified lead-based QDs with good dispersibility in aqueous and saline solutions. However, many of these studies did not fully address the complex interplay of conditions encountered in actual oil reservoirs, where tracers must perform reliably in the presence of crude oil, formation water, and fluctuating temperatures and pressures. (Morelli et al. 2012, Pichaandi et al. 2013)

To bridge these knowledge gaps and address the technological needs for advanced tracer systems, this work presents a detailed investigation into a core/shell quantum dot system engineered explicitly for oil reservoir tracing. We focus on core/shell nanocrystals featuring organic ligands, designed for optimal performance in oil-phase applications. These nanostructures offer significant advantages over conventional tracer technologies, including the potential for precisely tunable emission wavelengths for enhanced detection sensitivity, exceptional stability in the presence of crude oils with varying compositions, remarkable resilience in aqueous and high-salinity environments, consistent performance under reservoir-relevant temperature and pressure conditions, and high detection sensitivity at parts-per-billion concentrations. This study systematically evaluates these QD tracers under a comprehensive suite of conditions pertinent to oil reservoir environments, demonstrating their potential to overcome the limitations of traditional tracers and provide a powerful new tool for monitoring and managing oil reservoir dynamics with enhanced precision and reliability.

Experimental Section

Materials

All chemicals were procured from commercial suppliers and utilized as received without additional purification. These included anhydrous acetone (Amresco), cadmium oxide (CdO, 99.95%), zinc acetate dihydrate (98%, Acros Organics), anhydrous ethanol (BeanTown Chemical), hexane (ACS grade, Thermo Scientific), 1-octadecene (ODE, technical grade, 90%), octane (98%), 1-octanethiol (97%), oleic acid (OA, technical grade, 90%), oleylamine (OLAM, technical grade, 70%), dioctylamine (DOA, 97%), rhodamine 101 inner salt (R101, 94%, Thermo Scientific), selenium powder (Se, 99.5%, Thermo Scientific), sulfur powder (S, 99.999%, Thermo Scientific), toluene (99.8%), and tri-n-octylphosphine (TOP, 97%). Crude oil samples were supplied by Aramco and used directly or after dilution with toluene or n-hexane as required.

Synthesis of CdS/CdSe/CdS/ZnS Core/Shell Quantum Dots. The synthesis of the multi-shell QDs involved several sequential steps, adapting previously established colloidal synthesis methodologies. (Cassidy et al. 2020)

Initial CdS Nanocrystal Core Synthesis (6-9 nm): Bulk-sized CdS nanocrystals (NCs) were prepared following a coalescence-based growth procedure. Briefly, 42 mg of CdCl₂ was dissolved in 8 mL of OLAM under an argon atmosphere and heated to 240 °C. A suspension of small CdS seed NCs (2-4 nm diameter, 540 nmols) was rapidly injected, and the mixture was allowed to coalesce for 60 minutes. After quenching and cleaning via precipitation (toluene/ethanol), the NCs were redissolved in ODE and OA, heated to 150 °C for 60 minutes for surface treatment, then cleaned again and stored in hexane. Larger CdS cores (10-12 nm) were synthesized similarly but at 270 °C with 1080 nmols of seeds.

CdSe Emissive Shell Growth (CdS/CdSe): A CdSe shell was grown onto the CdS cores using syringe pump injection of 0.1 M Cd-oleate and 0.1 M TOP-Se precursors (prepared separately). CdS NCs (540 or 1080 nmols) were dispersed in DOA and ODE, degassed, and heated to 315 °C under argon. Precursor injection commenced at 270 °C at a rate of 3 mL/hr until the desired emission wavelength (typically 630-680 nm) was achieved (approx. 70-90 minutes). The reaction was then cooled and the NCs cleaned.

Inner CdS Shell Growth (CdS/CdSe/CdS): An additional CdS shell was deposited using Cd(oleate)₂ and octanethiol precursors. The CdS/CdSe NCs were dispersed in DOA and ODE, degassed, and heated to 315 °C. Precursors were injected at 3 mL/hr starting at 270 °C for 3-4 hours. After injection, the solution was annealed for 45 minutes before cooling and cleaning.

Outer ZnS Protective Shell Growth (CdS/CdSe/CdS/ZnS): The final ZnS shell was formed via a ZnS-CdS alloying strategy using three precursors: 0.1 M TOP-S, 0.1 M Cd-Oleate, and 0.1 M Zn(OA)₂. CdS/CdSe/CdS NCs were dispersed in ODE and DOA, degassed, and heated to 260 °C. A sequence of injections with varying Cd: Zn precursor ratios (3:1, then 2:2, then 1:3 mL mixtures with S-TOP) was performed, each for 30 minutes at 3 mL/hr. A final injection of 100% Zn(OA)₂ and S-TOP was conducted for 40 minutes, during which the temperature was raised to 315 °C. After a final OA injection and slow cooling, the QDs were cleaned and dispersed in hexane.

Characterization Methods

Absorbance spectra were recorded using a Cary 60 scan spectrophotometer. Photoluminescence (PL) spectra were obtained by exciting samples with a 405 nm pulsed laser diode (PDL 800-D, Picoquant); emitted photons were collected and analyzed using an Andor Shamrock 303i spectrograph coupled with an Andor Newton 970 EMCCD. Transmission electron microscopy (TEM) and High-Angle Annular Dark-Field Scanning TEM (HAADF-STEM) images were captured on a Thermo Fisher Talos F200X G2 S/TEM operated at 200 kV, using carbon-coated copper grids. Relative photoluminescence quantum yields (PLQY) were determined using a 532 nm CW DPSS laser (MeshTel) with Rhodamine 101 dye in ethanol (QY 99%) as a reference. Absolute PLQY values were measured with a Quantaurus-QY C11347 spectrometer (Hamamatsu, Inc.).

SARA Analysis

The SARA (Saturates, Aromatics, Resins, Asphaltenes) analysis of crude oil samples was performed using an IATROSCAN MK7 (TLC-FID) system following standard procedures involving sequential solvent elution on Chromarods and flame ionization detection.

Results and Discussion

The synthesized quantum dots were designed to exhibit robust emissive properties suitable for oil reservoir tracing. Figure 1 presents a schematic of the core/shell QD architecture via the HAADF-STEM images (not shown herein), confirming the multi-shell structure and particle dimensions. The calculated radial

probability distributions for electron and hole wave functions indicate confinement within the emissive CdSe layer, which is crucial for efficient luminescence. TEM images further corroborate the core size (e.g., 7.2 nm or 9.0 nm for the initial CdS core) and the successful deposition of subsequent shell layers.

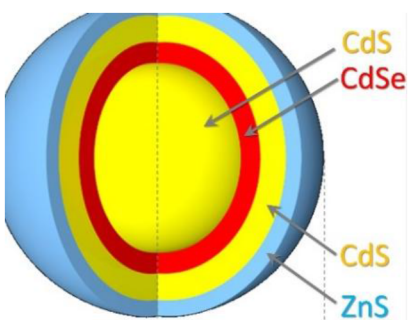


Figure 1—Schematic representations for the utilized QDs herein.

Optical characterization, as shown in Figure 2, reveals key features of the QDs and their interaction with crude oil. The QDs exhibit a broad absorption spectrum extending from 300 nm to 650 nm, while their emission is centered sharply around 650 nm, producing a strong red fluorescence. In contrast, crude oil displays a relatively featureless absorption spectrum (Figure 2). Upon addition of approximately 7.0 ppm of QDs to crude oil, the absorption spectrum shows minimal change. However, the 2D emission map (Figure 2) clearly distinguishes the sharp emission from the QDs at 650 nm from the broader, native fluorescence of the crude oil, demonstrating the potential for selective detection.

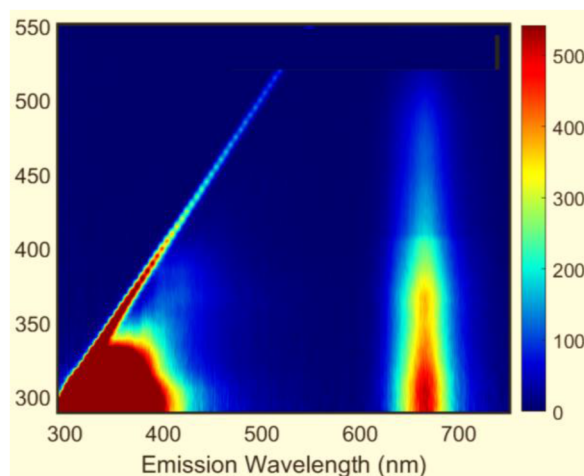


Figure 2—2D false steady state emission map for the emission from oil and the QDs at the both the blue and the red part of the spectrum.

To assess practical applicability, the detection limit of QDs in crude oil was determined. By adding varying concentrations (0.4-2.8 ppm) of QDs to crude oil samples, the lowest detection limit was estimated from emission data to be approximately 300 ppb. This sensitivity is vital for minimizing the amount of tracer required in field applications. A critical aspect for reservoir tracers is their stability in the presence of water. Experiments mixing QDs (1 ppm in crude oil) with increasing volumes of deionized water demonstrated remarkable stability; the QDs retained their emissive properties even when the water volume was 140,000 times that of the QD-oil mixture. This suggests a strong preference of the oleic acid-ligated QDs for the oil phase and their resistance to quenching or aggregation by water. Long-term stability tests (data not shown) over several months further confirmed the persistence of QD emission properties.

Reservoir formation water and injected water often contain high salt concentrations. Therefore, the stability of the QDs was evaluated in saline environments. The QDs maintain extraordinary emission stability in solutions containing 50,000 ppm and 200,000 ppm NaCl, even under continuous light irradiation for 60 minutes. Both absorption and emission spectra remained largely unchanged, highlighting their suitability for high-salinity reservoirs. The QDs were tested with seven different crude oil samples sourced from various fields, each exhibiting distinct SARA compositions. The QD tracer performance is largely independent of the bulk crude oil composition, a significant advantage for deployment across diverse oil fields.

Finally, the influence of reservoir-relevant temperatures and pressures on QD emission was investigated. The QDs exhibited robust emission characteristics when subjected to temperatures ranging from 30°C to 70°C and pressures from atmospheric up to 2000 psi. Only minor changes in emission intensity were observed, potentially attributable to some ligand desorption at the more extreme conditions, suggesting that further optimization of surface passivating ligands could enhance stability even further. These findings confirm the QDs' resilience under conditions mimicking those found downhole.

Conclusions

This study successfully developed and characterized a novel core/shell quantum dot (QD) system, specifically CdS/CdSe/CdS/ZnS nanocrystals, for application as fluorescent tracers in oil reservoir monitoring. The key findings demonstrate their significant potential to overcome limitations associated with conventional tracer technologies:

1. **Robust Synthesis and Characterization:** The multi-shell QDs were synthesized with a structure designed for high stability and efficient red-light emission (centered at ~650 nm), which is clearly distinguishable from the native fluorescence of crude oil.
2. **High Sensitivity:** The QDs exhibit a low detection limit of approximately 300 ppb in crude oil, enabling their use in minimal concentrations for effective tracing.
3. **Exceptional Stability in Harsh Conditions:** The tracers demonstrated remarkable stability in the presence of excessive volumes of deionized water (up to 140,000:1 water-to-oil mixture volume) and in high-salinity environments (50,000 ppm to 200,000 ppm NaCl), conditions commonly encountered in petroleum reservoirs.
4. **Versatility Across Crude Oil Types:** The emission properties of the QDs remained consistent across seven different crude oil samples with varying SARA compositions, indicating broad applicability irrespective of the specific oil chemistry.
5. **Resilience under Reservoir Temperatures and Pressures:** The QDs maintained their fluorescent characteristics with only minor changes under temperatures ranging from 30°C to 70°C and pressures up to 2000 psi, confirming their suitability for downhole applications.

In summary, the engineered QDs represent a significant advancement in tracer technology. Their exceptional stability, distinct emissive properties, and high sensitivity offer a robust and reliable tool for monitoring fluid dynamics, assessing well connectivity, and characterizing reservoir behavior with greater precision. These attributes pave the way for improved reservoir management and enhanced oil recovery strategies.

Acknowledgements

(Placeholder: The authors would like to acknowledge [Aramco/project number CIPR2375] for their support of this research.)

Nomenclature

CQD	= Carbon Quantum Dot
DOA	= Dioctylamine
HAADF-STEM	= High-Angle Annular Dark-Field Scanning Transmission Electron Microscopy
NCs	= Nanocrystals
ODE	= 1-Octadecene
OLAM	= Oleylamine
PLQY	= Photoluminescence Quantum Yield
QD	= Quantum Dot
SARA	= Saturates, Aromatics, Resins, Asphaltenes
SPE	= Society of Petroleum Engineers
TEM	= Transmission Electron Microscopy
TOP	= Tri-n-octylphosphine

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